

Week 2 Module

① find student Roll number (linear Search)

```
#include <stdio.h>
int main () {
    int N, x, found = 0;
    scanf ("%d", &N);
    int roll_numbers[N];
    for (int i = 0; i < N; i++) {
        scanf ("%d", &roll_numbers[i]);
    }
    scanf ("%d", &x);
    for (int i = 0; i < N; i++) {
        if (roll_numbers[i] == x) {
            printf ("found at index %d\n", i);
            found = 1;
            break;
        }
    }
    if (!found) {
        printf ("Not found\n");
    }
    return 0;
}
```

Input :- 5

101 105 113 110 130

O/P :- 110
found at index 3

#ThereIsOnlyOne

② Search product ID (Binary Search)

```
#include <stdio.h>
int main () {
    int N, target, found = 0;
    scanf ("%d", &N);
    int product_ids[N];
    for (int i = 0; i < N; i++) {
        scanf ("%d", &product_ids[i]);
    }
    scanf ("%d", &target);
    int low = 0;
    int high = N - 1;
    while (low <= high) {
        int mid = low + (high - low) / 2;
        if (product_ids[mid] == target) {
            printf ("product Available\n");
            found = 1;
            break;
        } else if (product_ids[mid] < target) {
            low = mid + 1;
        } else {
            high = mid - 1;
        }
    }
    if (!found) {
        printf ("product not Available\n");
    }
}
```

return 0;
}

I/p :- 6

10 20 30 40 50 60
40

O/p :- product Available

③ Arrange Exam Scores (Bubble Sort)

```
#include <stdio.h>
int main() {
    int N;
    scanf("%d", &N);
    int scores[N];
    for(int i=0; i<N; i++) {
        scanf("%d", &scores[i]);
    }
    for(int i=0; i<N-1; i++) {
        for(int j=0; j<N-i-1; j++) {
            if(scores[j] > scores[j+1]) {
                scores[j] = scores[j+1];
                scores[j+1] = temp;
            }
        }
    }
    for(int i=0; i<N; i++) {
        printf("%d", scores[i]);
        if(i<N-1) {
            printf(" ");
        }
    }
    printf("\n");
    return 0;
}
```

I/P :- 5
89 45 78 12 99

O/P :- 12 45 78 89 99

④ Arrange Library Book Numbers (Insertion Sort)

```
#include <stdio.h>
int main () {
    int N;
    scanf ("%d", &N);
    int books [N];
    for (int i = 0; i < N; i++) {
        scanf ("%d", &books [i]);
    }
    for (int i = 1; i < N; i++) {
        int key = books [i];
        int j = i - 1;
        while (j >= 0 & books [j] > key) {
            books [j + 1] = books [j];
            j--;
        }
        books [j + 1] = key;
    }
    for (int i = 0; i < N; i++) {
        printf ("%d", books [i]);
        if (i < N - 1) {
            printf (" ");
        }
    }
    printf ("\n");
    return 0;
}
```

I/P: 5
55 23 87 11 34

O/P: -
11 23 34 55 87

⑤ Rank players by Score (Selection Sort)

```
#include <stdio.h>
int main () {
    int N;
    scanf ("%d", &N);
    int Scores[N];
    for (int i=0; i<N; i++) {
        scanf ("%d", &Scores[i]);
    }
    for (int i=0; i<N-1; i++) {
        int min_idx = i;
        for (int j=i+1; j<N; j++) {
            if (Scores[j] < Scores[min_idx])
                min_idx = j;
        }
        int temp = Scores[min_idx];
        Scores[min_idx] = Scores[i];
        Scores[i] = temp;
    }
    for (int i=0; i<N; i++) {
        printf ("%d", Scores[i]);
        if (i < N-1)
            printf (" ");
    }
    printf ("\n");
    return 0;
}
```

IP:-
5
70 20 90 10 40
O/P:-
10 20 40 70 90

#ThereIsOnlyOne

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Language C

```
main.c
1 #include <stdio.h>
2
3 int main() {
4     int N, X, found = 0;
5
6     scanf("%d", &N);
7
8     int roll_numbers[N];
9     for (int i = 0; i < N; i++) {
10         scanf("%d", &roll_numbers[i]);
11     }
12
13     scanf("%d", &X);
14
15     for (int i = 0; i < N; i++) {
16         if (roll_numbers[i] == X) {
17             printf("Found at index %d\n", i);
18             found = 1;
19             break;
20         }
21     }
22
23     if (!found) {
24         printf("Not Found\n");
25     }
26
27     return 0;
28 }
```

Input

101 105 113 110 130
110
Found at index 3

...Program finished with exit code 0
Press ENTER to exit console.

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Language C

```
main.c
1 #include <stdio.h>
2
3 int main() {
4     int N, target, found = 0;
5
6     scanf("%d", &N);
7
8     int product_ids[N];
9     for (int i = 0; i < N; i++) {
10         scanf("%d", &product_ids[i]);
11     }
12
13     scanf("%d", &target);
14
15     int low = 0;
16     int high = N - 1;
17
18     while (low <= high) {
19         int mid = low + (high - low) / 2;
20
21         if (product_ids[mid] == target) {
22             printf("Product Available\n");
23             found = 1;
24             break;
25         } else if (product_ids[mid] < target) {
26             low = mid + 1;
27         } else {
28             high = mid - 1;
29         }
30     }
31
32     if (!found) {
33         printf("Product Not Available\n");
34     }
35
36     return 0;
37 }
```

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Language C

```
main.c
1 #include <stdio.h>
2
3 int main() {
4     int N;
5     scanf("%d", &N);
6
7     int scores[N];
8     for (int i = 0; i < N; i++) {
9         scanf("%d", &scores[i]);
10    }
11
12    for (int i = 0; i < N - 1; i++) {
13        for (int j = 0; j < N - i - 1; j++) {
14            if (scores[j] > scores[j + 1]) {
15                int temp = scores[j];
16                scores[j] = scores[j + 1];
17                scores[j + 1] = temp;
18            }
19        }
20    }
21
22    for (int i = 0; i < N; i++) {
23        printf("%d", scores[i]);
24        if (i < N - 1) {
25            printf(" ");
26        }
27    }
28    printf("\n");
29
30    return 0;
31 }
```

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main.c

```
1 #include <stdio.h>
2
3 int main() {
4     int N;
5     scanf("%d", &N);
6
7     int scores[N];
8     for (int i = 0; i < N; i++) {
9         scanf("%d", &scores[i]);
10    }
11
12    for (int i = 0; i < N - 1; i++) {
13        for (int j = 0; j < N - i - 1; j++) {
14            if (scores[j] > scores[j + 1]) {
15                int temp = scores[j];
16                scores[j] = scores[j + 1];
17                scores[j + 1] = temp;
18            }
19        }
20    }
21
22    for (int i = 0; i < N; i++) {
23        printf("%d", scores[i]);
24    }
```

input

```
5
89 45 78 12 99
12 45 78 89 99

...Program finished with exit code 0
Press ENTER to exit console.
```

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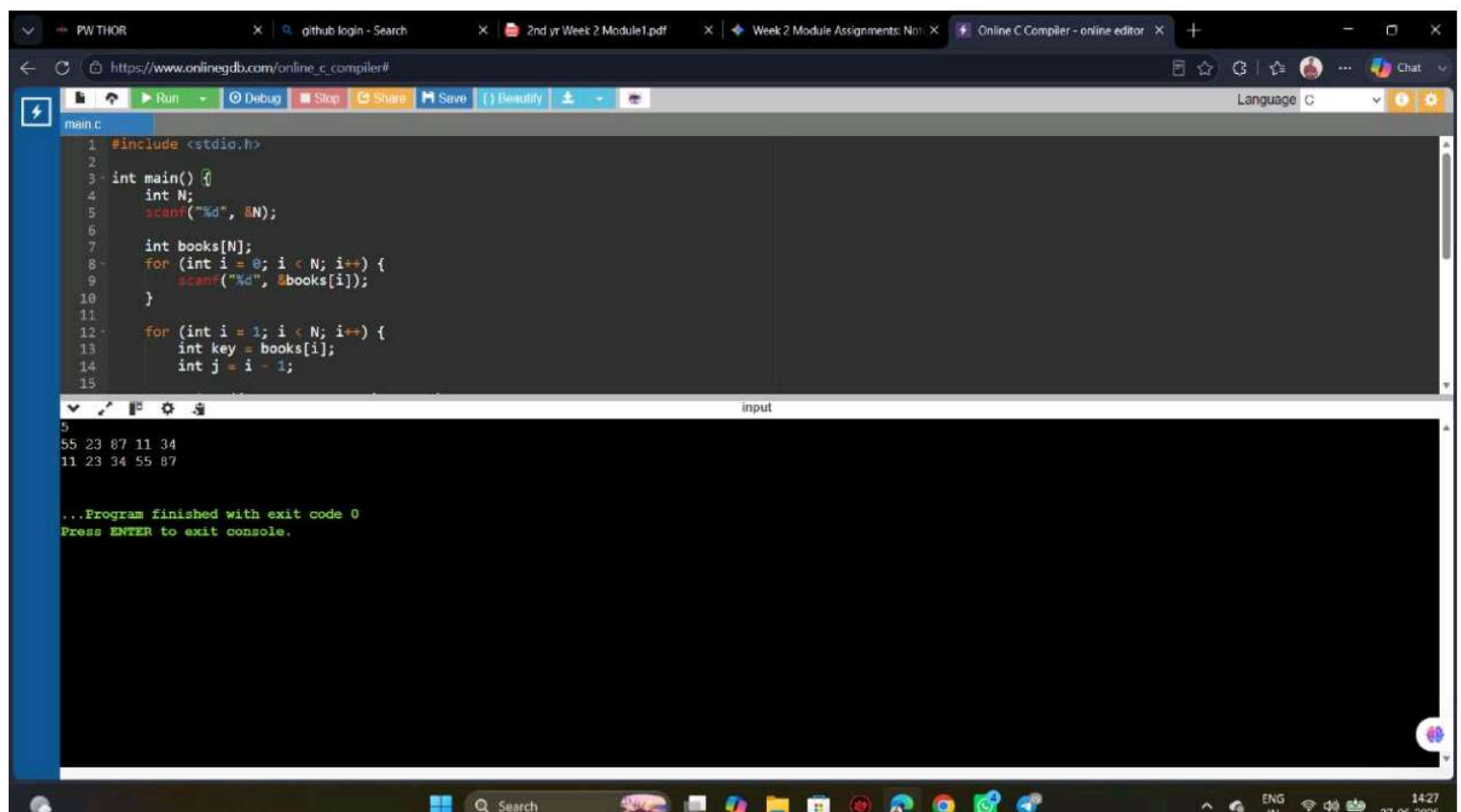
main.c

```
1 #include <stdio.h>
2
3 int main() {
4     int N;
5     scanf("%d", &N);
6
7     int books[N];
8     for (int i = 0; i < N; i++) {
9         scanf("%d", &books[i]);
10    }
11
12    for (int i = 1; i < N; i++) {
13        int key = books[i];
14        int j = i - 1;
15
16        while (j >= 0 && books[j] > key) {
17            books[j + 1] = books[j];
18            j--;
19        }
20        books[j + 1] = key;
21    }
22
23    for (int i = 0; i < N; i++) {
24        printf("%d", books[i]);
25        if (i < N - 1) {
26            printf(" ");
27        }
28    }
29    printf("\n");
30
31    return 0;
32 }
```

Language C

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Language C

main.c

```
1 #include <stdio.h>
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3 int main() {
4     int N;
5     scanf("%d", &N);
6
7     int scores[N];
8     for (int i = 0; i < N; i++) {
9         scanf("%d", &scores[i]);
10    }
11
12    for (int i = 0; i < N - 1; i++) {
13        int min_idx = i;
14        for (int j = i + 1; j < N; j++) {
15            if (scores[j] < scores[min_idx]) {
16                min_idx = j;
17            }
18        }
19        int temp = scores[min_idx];
20        scores[min_idx] = scores[i];
21        scores[i] = temp;
22    }
23
24    for (int i = 0; i < N; i++) {
25        printf("%d", scores[i]);
26        if (i < N - 1) {
27            printf(" ");
28        }
29    }
30    printf("\n");
31
32    return 0;
33 }
```

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Language C

```
main.c
1 #include <stdio.h>
2
3 int main() {
4     int N;
5     scanf("%d", &N);
6
7     int scores[N];
8     for (int i = 0; i < N; i++) {
9         scanf("%d", &scores[i]);
10    }
11
12    for (int i = 0; i < N - 1; i++) {
13        int min_idx = i;
14        for (int j = i + 1; j < N; j++) {
15            if (scores[j] < scores[min_idx]) {
16                min_idx = j;
17            }
18        }
19    }
20
21    for (int i = 0; i < N; i++) {
22        printf("%d ", scores[i]);
23    }
24    printf("\n");
25}
```

Input

```
5
70 20 90 10 40
10 20 40 70 90

...Program finished with exit code 0
Press ENTER to exit console.
```

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